Assignment 02: Building a Bayes Classifier Using GMM Initialized by K-means Clustering

Objective:

The aim of this assignment is to build a Bayes Classifier using a Gaussian Mixture Model (GMM), where the parameters (mean, covariance matrix, and mixing coefficients) are initialized using the K-means clustering algorithm. The classifier will be trained on image datasets and used to classify test images into distinct categories.

1. Introduction:

Dataset 2(b): Real-world Data

This includes:

- **3-Class Scene images dataset**: This dataset includes **Training** & **Testing Dataset,** In each this dataset, it has **three classes** namely **botanical_garden**, **bus_interior** and **elevator_shaft**, each of this class having **50 samples.**so in total **150 samples** are there in **Training dataset** and **150 samples** are there in **Testing Dataset.**

- Where each image or samples from each class of training dataset is divided into **patches** and then extract the feature from these patches of an image. there are two different ways of representation for this process of feature extraction as given below:
    - **24-dimensional colour histogram vectors:**
        1. Consider **32 x 32 non overlapping patches** on every image (from training and test sets). For example, if the image size is 256 x 256, there will be 32 numbers of 32 x 32 non overlapping patches.
        2. Extract 8-bin colour histogram from every colour channel (**R, G** and **B**) from a patch. It results in 3, 8-dimensional feature vectors. Concatenate them to form a **24-dimensional feature vector.**
        3. Similarly extract 24-dimensional feature vectors from every patch.
        4. Stack the 24-dimensional feature vectors corresponding to every patch in an image and save them as a file in the corresponding class folder.
        5. Thus, an image is represented as set (collection) of 24-dimensional colour histogram vectors representation
        6. Repeat the above steps to all the images in training and test sets of all the classes.

- Colour Histogram computed as follows from a colour channel : **Note that colour histogram is computed at every patch.**
    - When the given image is read, it will be read as a 3-dimensional matrix of pixel values. Each patch of an image is also a 3-dimensional matrix of pixel values. Each dimension is corresponding to a colour channel. The pixel values in each colour channel are in the range 0 to 255.

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- **For a colour channel:**
    1. Divide this range into 8 equal bins.
    2. Count the number of pixels falling into each bin. This results in a vector of 8 values.
    3. This is the 8-dimensional colour histogram (from a colour channel) feature vector.
    4. Normalise this vector by dividing it by the number of pixels in that patch.

    5. Do the same for other colour channels. Concatenate those three 8-dimensional colour histogram vectors to form a 24-dimensional vector.

- **32-dimensional Bag of Visual Words (BoVW) feature vectors:**
    1. Take the 24-dimensional colour histogram feature vectors of all the training examples of all the classes.
    2. Group them into 32 clusters using K-means clustering algorithms.
    3. Now take an image, assign each 24-dimensional colour histogram feature vector to a cluster.
    4. Count the number of feature vectors assigned to each of the 32 clusters.
    5. This results in a 32-dimensional BoVW representation for that image.
    6. Normalise this vector by dividing it by the number of 24-dimensional histogram feature vectors in that image.
    7. Repeat this for every image in the training and test set.

2. Implementation:

1.K-means Clustering Algorithm

The **K-means** clustering algorithm follows these steps:

a. *Cluster Assignment (Similarity Calculation)*

The **cluster assignment step** is based on the **Euclidean distance** between each data point $\mathbf{x}_n$ and the current centroid $\boldsymbol{\mu}_k$ of the k-th cluster. The distance is computed as:

$$\text{distance}(\mathbf{x}_n, \boldsymbol{\mu}_k) = \| \mathbf{x}_n - \boldsymbol{\mu}_k \|^2 = (\mathbf{x}_n - \boldsymbol{\mu}_k)^T (\mathbf{x}_n - \boldsymbol{\mu}_k)$$

Where:

- $\mathbf{x}_n$ is the n-th data point.
- $\boldsymbol{\mu}_k$ is the centroid of the k-th cluster.

Each data point is assigned to the cluster with the **minimum distance**.

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b. *Updating Centroids*

After assigning each data point to a cluster, the **new centroid** $\boldsymbol{\mu}_k$ for the k-th cluster is updated as the **mean** of all points assigned to that cluster:

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{\mathbf{x}_n \in C_k} \mathbf{x}_n$$

Where:

- N_k is the number of points assigned to the k-th cluster.
- C_k is the set of all points assigned to the k-th cluster.

c. *Covariance Matrix Update*

The covariance matrix $\boldsymbol{\Sigma}_k$ for each cluster is calculated as:

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{\mathbf{x}_n \in C_k} (\mathbf{x}_n - \boldsymbol{\mu}_k)\,(\mathbf{x}_n - \boldsymbol{\mu}_k)^T$$

d. *Convergence Criterion*

The algorithm iterates between the cluster assignment and centroid update steps until the centroids no longer change significantly (i.e., the change in centroids is smaller than a threshold).

2. GMM (Gaussian Mixture Model) - EM Algorithm

The **Expectation-Maximization (EM)** algorithm is used to train the **Gaussian Mixture Model (GMM)**, which involves the following steps:

a. *Gaussian Probability Density Function (PDF)*

The formula used to compute the likelihood of a sample $\mathbf{x}_n$ under a Gaussian distribution $\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is given by:

$$p(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) = \frac{1}{(2\pi)^{d/2} \mid \boldsymbol{\Sigma}_k \mid^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}_n - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k)\right)$$

Where:

- d is the dimensionality of the data.
- $\mathbf{x}_n$ is the n-th data point.
- $\boldsymbol{\mu}_k$ is the mean of the k-th Gaussian component.
- $\boldsymbol{\Sigma}_k$ is the covariance matrix of the k-th Gaussian component.
- $\mid \boldsymbol{\Sigma}_k \mid$ is the determinant of the covariance matrix.
- $\boldsymbol{\Sigma}_k^{-1}$ is the inverse of the covariance matrix.

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b. *Responsibilities (Expectation Step)*

The **responsibility** γ_{nk} (probability that the n-th sample $\mathbf{x}_n$ belongs to the k-th Gaussian component) is computed in the **Expectation step (E-step)** as:

$$\gamma_{nk} = \frac{\pi_k \, \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \; \mathcal{N}\left(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j\right)}$$

Where:

- π_k is the mixing coefficient (weight) for the k-th Gaussian component.
- $\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the Gaussian PDF as defined above.
- The denominator is the sum of Gaussian PDFs over all K components.

c. *Updating Means (Maximization Step)*

The updated **mean** $\boldsymbol{\mu}_k$ for each Gaussian component is calculated using the responsibilities in the **Maximization step (M-step)**:

$$\boldsymbol{\mu}_k = \frac{\sum_{n=1}^{N} \gamma_{nk} \, \mathbf{x}_n}{\sum_{n=1}^{N} \gamma_{nk}}$$

Where:

- γ_{nk} is the responsibility that sample $\mathbf{x}_n$ belongs to the k-th Gaussian component.
- N is the total number of data points.

d. *Updating Covariances (Maximization Step)*

The updated **covariance matrix** $\boldsymbol{\Sigma}_k$ for each Gaussian component is:

$$\boldsymbol{\Sigma}_k = \frac{\sum_{n=1}^{N} \gamma_{nk} \, (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T}{\sum_{n=1}^{N} \gamma_{nk}}$$

Where:

- γ_{nk} is the responsibility of the n-th sample for the k-th Gaussian component.
- $\mathbf{x}_n - \boldsymbol{\mu}_k$ is the difference between the sample and the updated mean for the k-th component.

e. *Updating Weights (Mixing Coefficients)*

The updated **mixing coefficient** (weight) π_k for the k-th Gaussian component is:

$$\pi_k = \frac{1}{N} \sum_{n=1}^{N} \gamma_{nk}$$

Where:

- N is the total number of data points.
- γ_{nk} is the responsibility of the n-th sample for the k-th Gaussian component.

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f. Log-Likelihood Calculation

The **log-likelihood** $\mathcal{L}$ is computed at each iteration to track the convergence of the EM algorithm:

$$\mathcal{L} = \sum_{n=1}^{N} \log\left(\sum_{k=1}^{K} \pi_k \ \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\right)$$

Where:

- N is the number of data points.
- K is the number of Gaussian components.

3.Experimental Results: When Number of Component/Cluster is K=1

❖ Iteration Vs Log-Likelihood Graph:

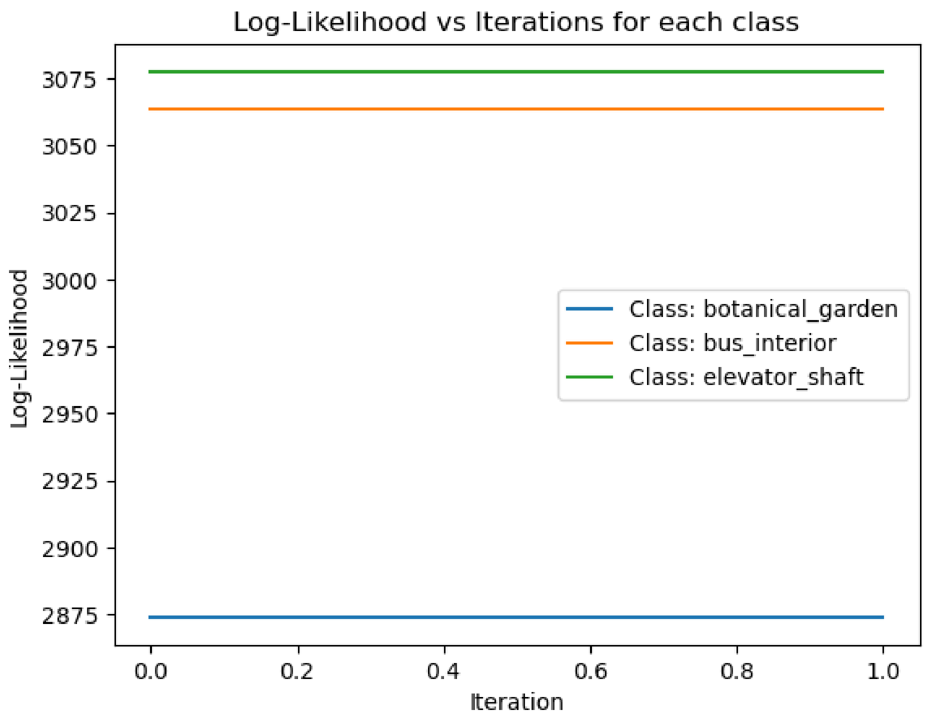

Figure 1

Flat Log-Likelihood Curves for All Classes:

- All three curves (for **Botanical Garden**, **Bus Interior**, and **Elevator Shaft**) are **flat**, which means that the log-likelihood values did **not change** across iterations.

- This implies that the algorithm **converged immediately** after the first iteration or did not proceed with iterative updates, which is expected when we use **only one cluster (K=1)**.
- **Inference**: With just **one cluster**, the initial model likely represented the entire data distribution quite well, and there was no need for additional iterations to improve the log-likelihood.
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4.Performance Matrix:

- **Accuracy:** 62.00 %
- **Confusion Matrix:**

Predicted	**Actual**		
	Class 1	Class 2	Class 3
Class 1	36	8	6
Class 2	6	29	15
Class 3	9	13	28

- **Class-Wise Precision, Recall, F1-Score:**
  - **botanical_garden Class 1:**
    1) Precision: 0.7059
    2) Recall: 0.7200
    3) F1-Score: 0.7129
  - **bus_Interior Class 2:**
    1) Precision: 0.5800
    2) Recall:0.5800
    3) F1-Score:0.5800
  - **elevator_shaft Class 3:**
    1) Precision:0.5714
    2) Recall:0.5600
    3) F1-Score:0.5657
- **Mean Precision, Recall, F1-Score (Weighted):**
  - Mean Precision :0.6191
  - Mean Recall: 0.6200
  - Mean F1-Score: 0.6195

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5. Conclusion

This project demonstrates the process of building a **Bayes Classifier** using **GMM**, where the parameters are initialized using **K-means clustering**. The classifier successfully learns the distribution of the data and classifies the test images with a high level of accuracy.

References

- Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.
- Murphy, K. P. (2012). Machine Learning: A Probabilistic Perspective. MIT Press.